Units of energy



J

joule (SI unit)

1 kJ = 1000 J

used in science



cal · kcal

calorie

1 cal = 4.18 J
1 kcal = 1000 cal

food labels: heat 1 g of water by 1 °C



kWh

kilowatt-hour

1 kWh = $3.6 \cdot 10^6$ J

electricity bills: a 1 kW device for 1 hour



toe

tonne of oil equivalent

1 toe ≈ $4.19 \cdot 10^{10}$ J

energy statistics: burning 1000 kg of oil